Assignment – 1.15

Hands-on Subnetting Practical

Q - 1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

- **Given Network:** 192.168.10.0/24

  We need to divide it into 4 equal subnets.

- **Subnet Mask:** 255.255.255.192 or /26

Subnet	Network Address	Broadcast Address
1	192.168.10.0	192.168.10.63
2	192.168.10.64	192.168.10.127
3	192.168.10.128	192.168.10.191
4	192.168.10.192	192.168.10.255

- **Result:** 4 equal subnets are created, each containing 64 IP addresses.

Q - 2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Hint: Make sure not to use the network or broadcast address for any device.

- **IP Addresses for 5 Devices:**

Selected subnet: 192.168.10.0/26

Subnet mask: 255.255.255.192

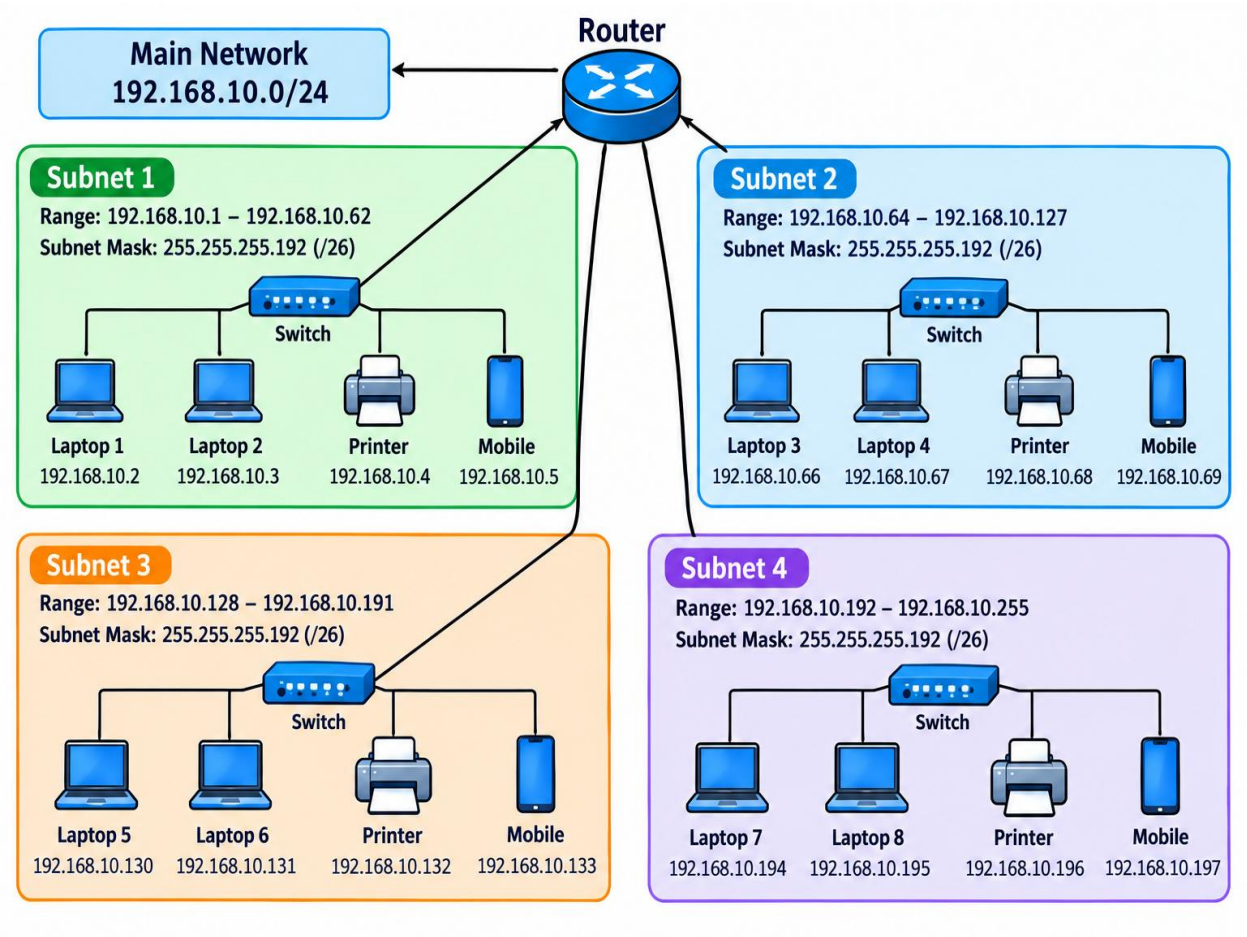
Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

Network address: 192.168.10.0

Broadcast address: 192.168.10.63

All five devices have valid IP addresses within the subnet.

Q - 3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



Q - 4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

- **Step-by-Step Subnetting:**
- **Step 1: Given Network:**

  Network address: 192.168.10.0/24

  We need to divide it into 4 equal subnets.

- **Step 2: Calculate Subnet Mask:**
  - New subnet mask: 255.255.255.192
  - Prefix: /26
  - We borrow 2 bits to create 4 subnets.
- **Step 3: Calculate Hosts:**
  - Total IPs per subnet = $2^6 = 64$
  - Usable IPs = 64 − 2 = **62 hosts**
- **Step 4: Four Subnets:**

Subnet	Network Address	Broadcast Address
1	192.168.10.0	192.168.10.63
2	192.168.10.64	192.168.10.127

Subnet	Network Address	Broadcast Address
3	192.168.10.128	192.168.10.191
4	192.168.10.192	192.168.10.255

- **Step 5: Assign IP Addresses:**

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5